

Option A: Immunology and Disease

7. Tuberculosis (TB) is the leading cause of death by an infectious disease worldwide.

(a) (i) State why tuberculosis is called an infectious disease. [1]

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(ii) To prevent the spread of TB, the BCG vaccination was introduced in 1953 for children aged 10–14 years old. Define the term vaccine. [1]

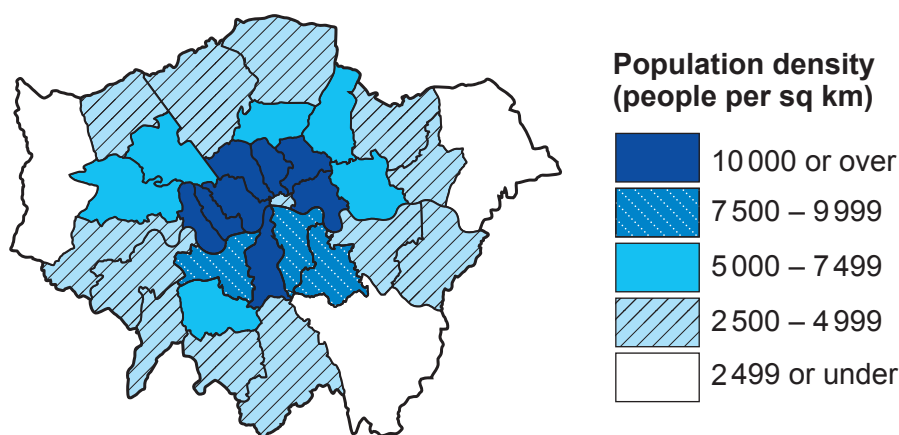
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The BCG vaccination is no longer part of childhood immunisations. It is only offered to those who are at higher risk of TB; for example, children under 5 who live in an area of the UK with a high rate of TB, as in some areas of London. **Image 7.1** shows the population density of different areas of London.

Image 7.1



(iii) Using the map in **Image 7.1**, suggest why the TB incidence is so high in some areas of London. [2]

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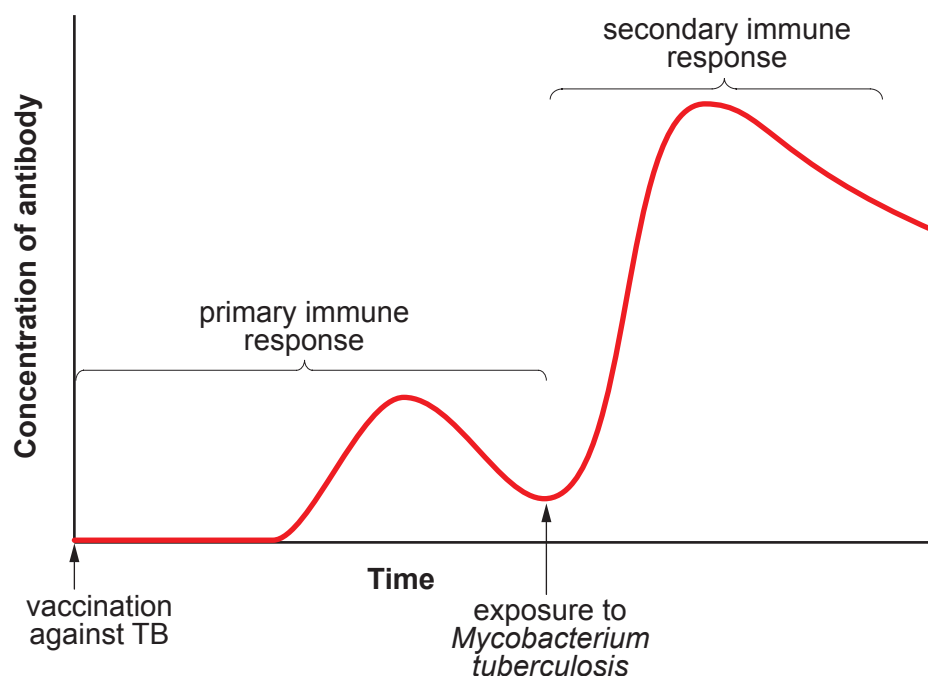
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Graph 7.2 shows the concentration of antibodies released by the body following a BCG vaccination and then after exposure to the live infectious micro-organism, *Mycobacterium tuberculosis*.

Graph 7.2



- (b) The BCG vaccine initiates an active immune response which leads to the production of antibodies by the patient.

- (i) Describe and explain the shape of the part of **Graph 7.2** which shows the primary immune response. [4]

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- (ii) With reference to **Graph 7.2**, explain how vaccination prevents development of TB when exposed to *M. tuberculosis*. [4]

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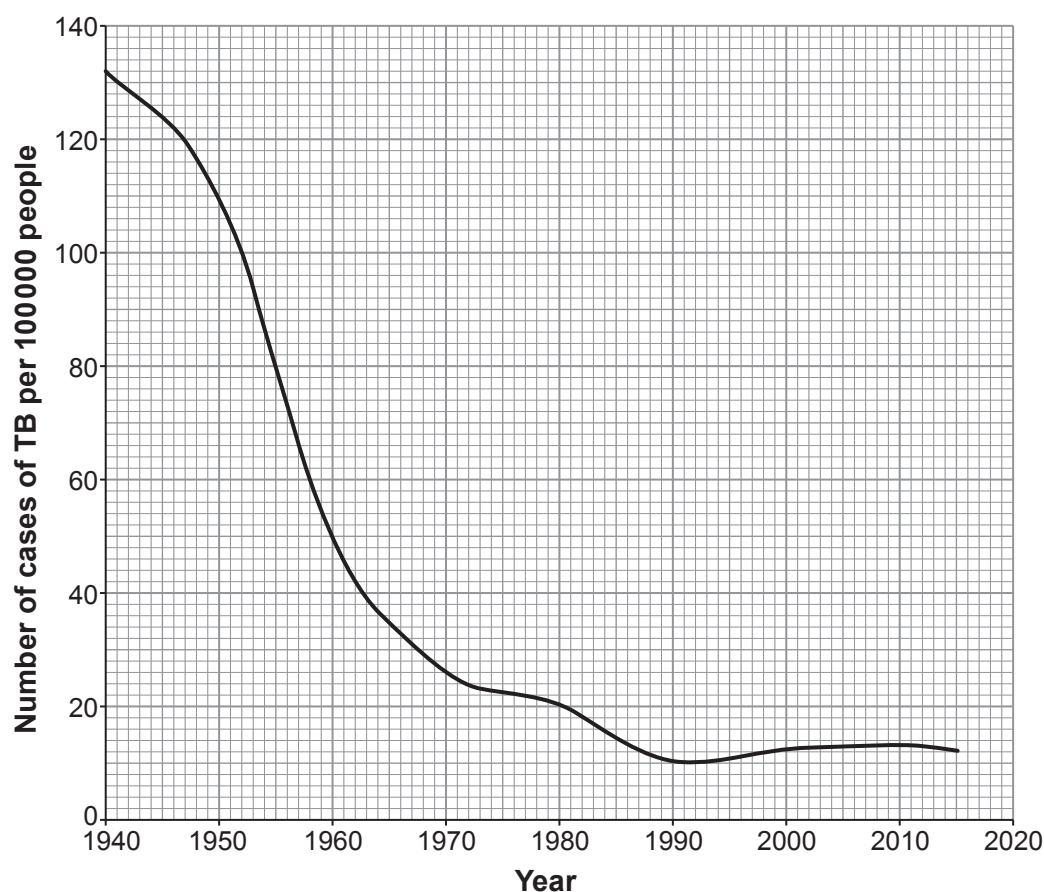
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- (c) Since the 1940s, antibiotics have been used to treat TB. **Graph 7.3** shows the change in the numbers of diagnosed TB cases in England and Wales from 1940 to 2015.

Graph 7.3



- (i) Calculate the percentage decrease in **TB incidence** between 1940 and 2015. [2]

Percentage decrease in incidence of TB = %



Mycobacterium tuberculosis can develop into a dormant state, referred to as latent TB. Whilst dormant, *M. tuberculosis* is resistant to antibiotics. Patients with TB receive a combination of antibiotics to kill active bacteria rapidly and to try to prevent latent TB.

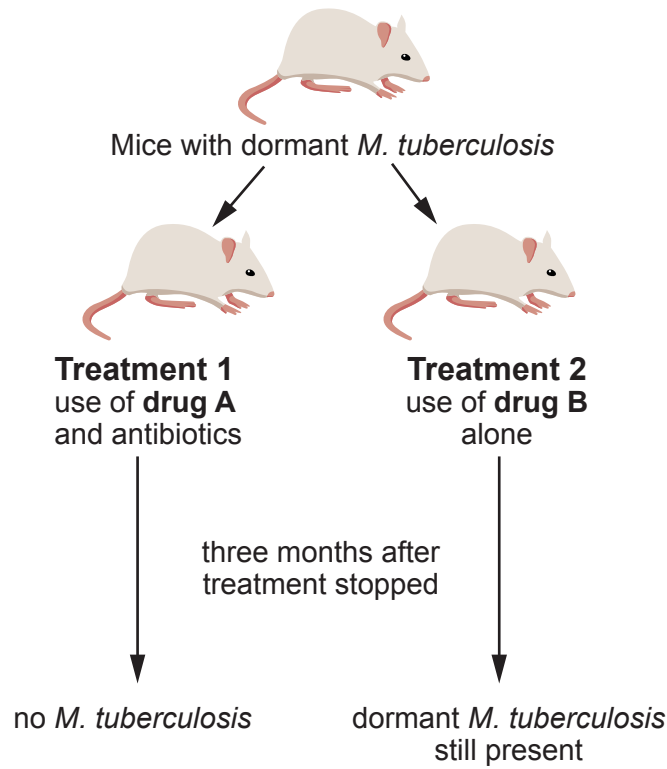
Two new treatments have been proposed to target dormant *M. tuberculosis* to prevent TB recurring in a patient.

Treatment 1
Use of **drug A** to activate the dormant bacteria as well as treating with high doses of antibiotics to kill all activated bacteria.

Treatment 2
Use of **drug B** that will inhibit the activation of any dormant bacteria while the drug is present.

These treatments have been trialled in mice as shown in **Image 7.4**.

Image 7.4



(ii) Complete the table below by stating **one** advantage and **one** disadvantage for each of the treatments. [4]

Treatment	Advantage	Disadvantage
Drug A and antibiotics		
Drug B alone		



- (d) The methods of treating TB can also be used to treat malarial infections.
Explain why these methods are less effective when they are used in the treatment of malarial infections. [2]

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